

# Amir Mirzai Golpayegani

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## Education

### M.Sc. in Computer Science

Sep 2022 – Feb 2026

University of Calgary, Calgary, AB, Canada

Relevant Courses: Computer Animation (A+), Modelling for Computer Graphics (A)

Thesis: Research in indexing schemes and continuous data representation in graphics and geospatial contexts

GPA: 3.940/4.0

### B.Sc. in Computer Engineering

Sep 2016 – Feb 2021

Sharif University of Technology, Tehran, Iran

GPA: 17.37/20 (WES: 3.69)

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## Publications

1. A. Mirzai Golpayegani, M. Hasan, and F. F. Samavati, “Real-time multiresolution management of spatiotemporal earth observation data using dggs,” *Remote Sensing*, vol. 17, no. 4, p. 570, 2025 [doi:10.3390/rs17040570](https://doi.org/10.3390/rs17040570)
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## Technical Skills

**Programming Languages:** Python, C++

**Frameworks & Libraries:** Django, TensorFlow, PyTorch, Scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Seaborn, IMGUI

**Technologies & Platforms:** AWS, Docker, PostgreSQL, OpenGL, GLSL, Linux, SNAP, RESTful APIs

**Tools & Software:** Git, Blender (3D Modeling), LaTeX

**Research & Specialized Skills:** Wavelet Transform (Signal Processing), Spatial Data Indexing, Data Visualization

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## Projects

### DEM Super-Resolution Framework

2024 – 2025

*Python, PyTorch, CNNs, Geospatial Data Processing, 3D Visualization, OpenGL*

- Co-developed a CNN-based deep learning framework to enhance Digital Elevation Models (DEMs) from 90 m to 30 m resolution while preserving high-frequency terrain details.
- Designed and implemented a global dataset collection pipeline to automatically gather DEM tiles worldwide.
- Built a 3D visualization tool to inspect and evaluate reconstructed high-resolution DEM outputs.

### Full-Stack Health & Wellness Tracker (Ongoing)

Jul 2025 – Present

[GitHub](#)

*Node.js, Express.js, PostgreSQL, JWT, Docker, GitHub Actions*

- Building a secure, scalable backend with user auth, activity logging, and AI-based diet recommendations.
  - Using JWT auth with bcrypt hashing and PostgreSQL for persistent data storage.
  - Containerizing with Docker and automating builds/deployment via GitHub Actions.
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## Work Experience

### Graphics Research Assistant

Sep 2022 – Feb 2026

*Intern @ [Biggeo](#), Calgary, Canada*

- Designed and implemented a C++ image-processing framework for large-scale Earth observation data, supporting multiresolution storage, compression, and real-time access.
- Built automated processing pipelines for multiresolution encoding of time-varying geospatial data by integrating wavelet-based spatial encoding and temporal approximation methods.
- Developed GPU-accelerated visualization and rendering components using OpenGL/GLSL.
- Extended an existing DGGS C++ codebase using GDAL, ensuring correctness, performance, and maintainability.
- Collaborated closely with researchers to translate algorithms into production-quality software with reproducible results.

### Back-End Developer

Jun 2020 – Jul 2022

*[Asr Gooyesh Pardaz](#), Tehran, Iran*

- Developed REST APIs in Python using Django, integrated with NLP tasks.
  - Implemented scalable backend solutions for speech recognition systems.
  - Diagnosed and resolved backend issues using logging, profiling, and unit testing to identify root causes and prevent recurrence.
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## Academic Experience

**Spatiotemporal Data Encoding and Storage Using Discrete Global Grids** Sep 2022 – Feb 2026  
*University of Calgary* Supervisor: [Dr. Faramarz Samavati](#)

- Efficient real-time storage of spatiotemporal Earth observation data.
- Developing a wavelet-based spatial representation of global Earth data to achieve compression and multiresolution storage.
- Using a B-spline-curve-based temporal representation of time-series data.

**Bachelor's Thesis: Persian Text Normalizer for NLP** Oct 2016 – Feb 2021  
*Sharif University of Technology* Supervisor: [Dr. Hossein Sameti](#)  
Developed a Python toolkit to standardize Persian texts before NLP processes.

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## Teaching Experience

**Teaching Assistant, Developing Big Data Applications** Dec 2024 – Apr 2025  
*University of Calgary* Instructor: [Dr. Tyler Bonnell](#)  
Mentored students in designing and deploying cloud-based solutions using AWS for large-scale data processing and storage, including hands-on guidance in debugging, optimizing workflows, and applying best practices for reliability and scalability.

**Teaching Assistant, Working with Data at Scale** Dec 2023 – Apr 2024  
*University of Calgary* Instructor: [Dr. Katie Ovens](#)  
Responsibilities included assisting with lectures, helping students with large-scale data processing techniques, and grading projects.

**Teaching Assistant, Database Management Systems** Dec 2022 – Apr 2023  
*University of Calgary* Instructor: [Dr. Reda Alhaj](#)  
Responsibilities included grading assignments and exams, and supporting students with SQL and database design concepts and mentoring them during their course projects.

**Teaching Assistant, Game Theory** Dec 2020 – Apr 2021  
*Sharif University of Technology* Instructor: [Dr. MohammadAmin Fazli](#)  
Responsibilities included holding tutorial sessions, grading assignments, and assisting students with course material.

**Teaching Assistant, Game Theory** Sep 2020 – Dec 2020  
*Sharif University of Technology* Instructor: [Dr. MohammadAmin Fazli](#)  
Responsibilities included conducting problem-solving sessions, and preparing course materials.

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## Languages

**English:** Full Professional Proficiency (TOEFL 115)  
**Persian:** Native Proficiency

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## Certificates

**Supervised Machine Learning: Regression and Classification** ([certificate](#)) Oct 2024  
*DeepLearning.AI and Stanford University, Coursera*

**Applied Machine Learning in Python** ([certificate](#)) Jul 2020  
*University of Michigan, Coursera*

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## Achievements

**Ranked in the Top 0.001% of Konkour Participants** May 2016  
*17th Nationwide, University Entrance Exam, Iran*